

Task 4: Data Cleaning & Reporting Automation

This project demonstrates automated data cleaning and reporting workflows using the Customer Personality Analysis dataset. The workflow includes handling missing values, removing duplicate records, standardizing data formats, generating summary statistics, and creating automated visual reports.

Objectives

- Handle missing values and inconsistent data
- Remove duplicate records
- Standardize categorical text fields
- Generate automated reports and visual summaries
- Improve reporting efficiency through automation

Tools Used

Python, Pandas, NumPy, Matplotlib, and Jupyter Notebook.

Data Cleaning Process

1. Loaded the Customer Personality Analysis dataset.
2. Identified missing values in the Income column.
3. Replaced missing Income values using the median value.
4. Checked and removed duplicate records where applicable.
5. Standardized text fields such as Education and Marital_Status.
6. Converted customer enrollment dates into datetime format.
7. Exported a cleaned dataset and generated summary statistics.

Visualizations

Figure 1: Missing Values After Cleaning

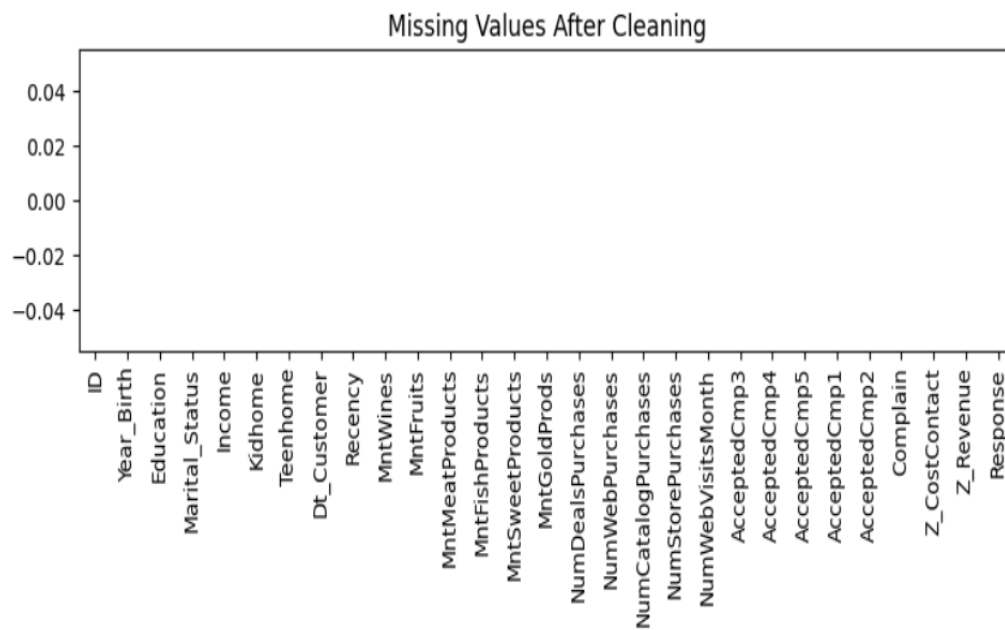


Figure 2: Education Distribution

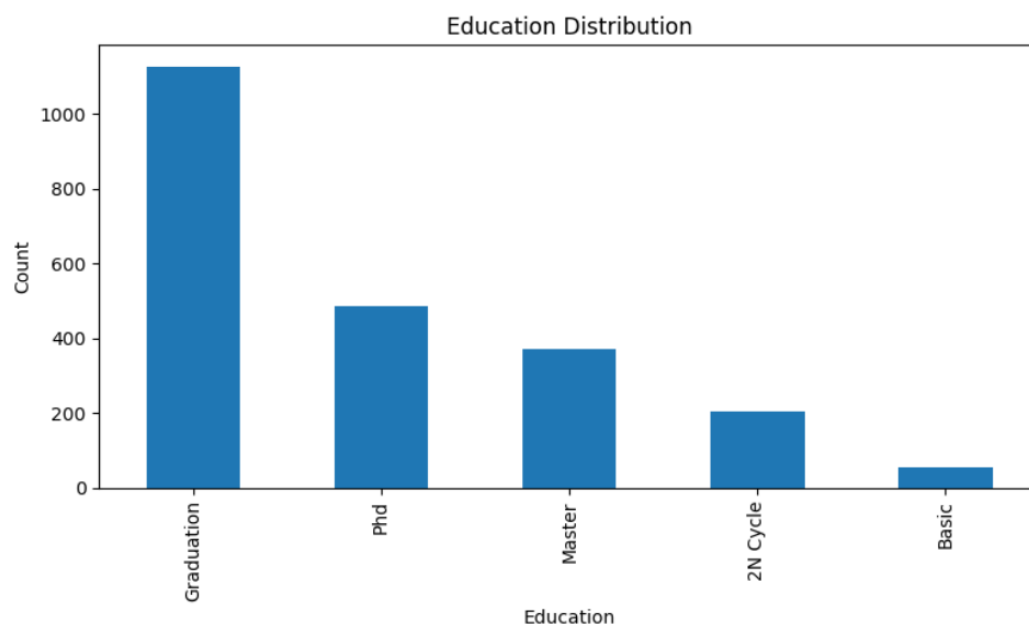
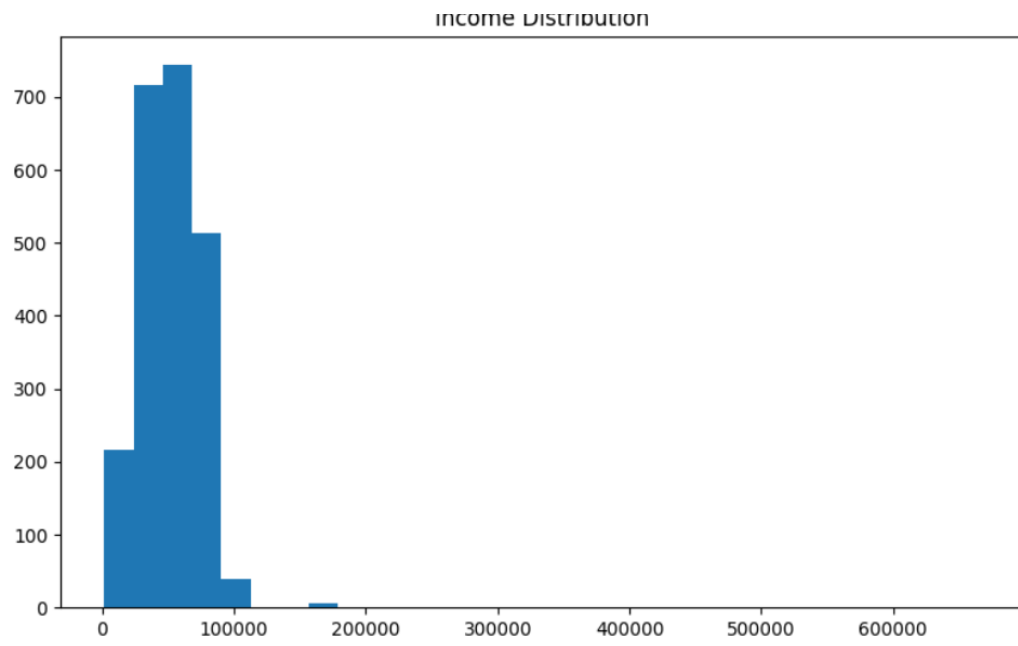


Figure 3: Income Distribution



Results

- Missing values were successfully handled.
- Cleaned dataset was generated automatically.
- Summary statistics were exported for reporting.
- Visualizations were created to support analysis.
- The automation workflow reduced manual preprocessing effort and improved data quality.

Conclusion

The Data Cleaning & Reporting Automation project successfully demonstrated practical data preprocessing techniques and automated reporting. The workflow ensured improved data quality, reproducibility, and efficiency while generating meaningful insights from the Customer Personality Analysis dataset.